

Exercises - Probability, Entropy and MC Simulation

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1 Basic Probabilities - Axiom and Propositions

Exercise 1 *A committee of 5 is to be selected from a group of 6 men and 9 women. If the selection is made randomly, what is the probability that the committee consists of 3 men and 2 women? If there are two men who are rival. What is the probability of forming such committee these two men are not in that committee?*

Exercise 2 *If you draw two cards without replacement, what is the probability of getting two Hearts? What is the probability of getting no hearts?*

Exercise 3 *Draw a card from deck of card. Find the probabilities of the following events:*

- (i) *drawing a face card (Jack, Queen, or King),*
- (ii) *drawing a even numbered card,*
- (iii) *drawing a odd numbered card, and*
- (iv) *drawing a black prime numbered card.*

Exercise 4 *Draw five cards from a deck.*

- (i) *What is the probability of getting a flush (all cards of the same suit)?*
- (ii) *What is the probability of drawing four same numbered cards?*

Exercise 5 *If you draw three cards without replacement, what's the probability of*

- (i) *getting three different suits,*
- (ii) *getting same suite in all three cards,*
- (iii) *getting 2 same suites?*

Exercise 6 If you draw five cards without replacement, what is the probability that at least one of them is an ace? What is the probability that, you get no ace?

Exercise 7 For any two events E and F of a sample space S , prove that

$$(i) P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

$$(ii) P(E \setminus F) = P(E) - P(E \cap F).$$

$$(iii) \text{ If } F \subseteq E, \text{ then prove that } P(E \setminus F) = P(E) - P(F) \text{ and } P(E) \geq P(F).$$

Exercise 8 Prove general law of addition of probabilities for any finite collection of events. That is, for any finite collection of events $\{E_i\}_{i=1}^n$, prove that

$$\begin{aligned} P\left(\bigcup_{i=1}^n E_i\right) &= \sum_{i=1}^n P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} \cap E_{i_2}) + \sum_{i_1 < i_2 < i_3} P(E_{i_1} \cap E_{i_2} \cap E_{i_3}) - \cdots \\ &\quad + (-1)^{r+1} \sum_{i_1 < i_2 < i_3} P(E_{i_1} \cap E_{i_2} \cap E_{i_3} \cap \cdots \cap E_{i_r}) \\ &\quad + \cdots + (-1)^{n+1} P(E_1 \cap E_2 \cap \cdots \cap E_n). \end{aligned} \quad (1)$$

Hint: Use mathematical induction.

Exercise 9 Two dice are thrown. Let E be the event of sum of outcome of two dice is odd and F be the event that at least one of the dice outcome is 1. Enumerate/describe the following events and calculate their probabilities.

1. Sample space,
2. $E \cap F$,
3. $E \cup F$,
4. $E \setminus F$, and
5. $F \setminus E$.

Exercise 10 A bowl contains 4 chits numbered 1, 2, 3 and 4 and another bowl contains 6 chits numbered 2, 4, 6, 7, 8 and 9. If one of the bowl is chosen at random and then a chit is drawn from the chosen bowl, find the probabilities that the chit drawn bears the number (i) 2 or 4, (ii) 3, (iii) 1 or 9.

Exercise 11 A card is drawn randomly from a deck of cards. What is the probability that the drawn card is a heart or a jack?

Hint: Use $P(E \cup F)$ formula.

Exercise 12 An urn contains 6 red, 4 white, and 5 black balls. Four balls are drawn at random. What is the probability that there is at least one red, one white and one black ball (that is, no colour is missed out).

Exercise 13 Let n be an even number. For some experiment, let the sample space be $S = \{1, \frac{1}{2}, \frac{1}{2^2}, \dots, \frac{1}{2^n}\}$. Let $E = \{1, \frac{1}{2}\}$ and $F = \{\frac{1}{2^i} : i \text{ is even}\}$. Find $P(E^c \cap F^c)$.

Exercise 14 (Birthday Problem) In a group of n people, what is the probability that no two persons have the same birthday.

Exercise 15 Consider the experiment. There are infinitely many balls numbered $b_1, b_2, b_3, \dots$. In a time interval of 0 to 1, we mark the points $0, 1 - \frac{1}{2}, 1 - \frac{1}{2^2}, 1 - \frac{1}{2^3}, \dots$. At each of these marked times, we put some set of balls in a urn and remove one ball out of it. Assume that both put and remove happen instantly (without any time delay).

Experiment 1: At time 0, we put the balls numbered $b_1, b_2, \dots, b_{10}$ and remove the ball numbered b_{10} . At the next time $1 - \frac{1}{2}$, we put the balls numbered $b_{11}, b_{12}, \dots, b_{20}$ and remove the ball numbered b_{20} . At the next time $1 - \frac{1}{2^2}$, we put the balls numbered $b_{21}, b_{22}, \dots, b_{30}$ and remove the ball numbered b_{30} . In general, at time $1 - \frac{1}{2^n}$, put the balls numbered $b_{10n+1}, b_{10n+2}, \dots, b_{10(n+1)}$ and remove the ball numbered $b_{10(n+1)}$. Following the pattern, continue this process infinitely. At time 1, how many number of balls are there in the urn and find the probability that the ball numbered

- (i) b_{200} is in the urn,
- (ii) b_{87} is in the urn,
- (iii) b_{2020} is in the urn, and
- (iv) $b_{10^{405}-1}$ is in the urn.

Experiment 2: At time 0, we put the balls numbered $b_1, b_2, \dots, b_{10}$ and remove the ball numbered b_1 . At the next time $1 - \frac{1}{2}$, we put the balls numbered $b_{11}, b_{12}, \dots, b_{20}$ and remove the ball numbered b_2 . At the next time $1 - \frac{1}{2^2}$, we put balls numbered $b_{21}, b_{22}, \dots, b_{30}$ and remove the ball numbered b_3 . In general, at time $1 - \frac{1}{2^n}$, put the balls numbered $b_{10n+1}, b_{10n+2}, \dots, b_{10(n+1)}$ and remove the ball numbered $b_{(n+1)}$. Following the pattern, continue this process infinitely. At time 1, how many number of balls are there in the urn and find the probability that the ball numbered

- (i) b_{200} is in the urn,
- (ii) b_{87} is in the urn,
- (iii) b_{2020} is in the urn, and
- (iv) $b_{10^{405}-1}$ is in the urn.

Exercise 16 Roll a fair die twice. What is the probability of the sum of two rolls equals 6 given that same number comes up on both the rolls.

Exercise 17 On rolling a dice twice, prove that the events below

- (i) first roll's outcome is 6 and second roll's outcome is 6,
 - (ii) Sum of the two rolls is 7 and second roll's outcome is 4,
- are independent.

Exercise 18 (Law of Total Probability) Let S be the sample space and let $F_1, F_2, \dots, F_n$ be mutually exclusive collection of events such that it partitions S , that is, $\bigcup_{i=1}^n F_i = S$. Then prove that, for any event E ,

$$P(E) = \sum_{i=1}^n P(E|F_i) P(F_i).$$

Exercise 19 Suppose that two dice are rolled. Note the events

- (i) E when second dice roll is 4.
 - (ii) F when the difference of two rolls is 4.
 - (iii) G when difference between the two rolls is 3.
- (a) Are E and F independent? (b) Are G and F independent?

Exercise 20 State and prove Baye's rule.